ST 6100 CID2 Installation Guide (Hazardous Location)

The ST 6100 CID2 terminals are hazardous location satellite communications terminals manufactured by ORBCOMM. ORBCOMM distributes the terminals through Solution Providers who activate the terminals on the ORBCOMM satellite network and provide application programs that communicate with the terminals.

This guide provides the necessary information to ensure successful installation of the terminals for Class I Division 2, Group A, B, C and D Hazardous Locations. This guide does not address the terminal's commissioning procedures as these procedures vary depending upon the Solution Provider and their specific application. The terminals are available with a mounting bracket suitable for applications in hazardous locations requiring certification to Class I Division 2, Group A, B, C and D locations.

WARNING ***This equipment is suitable for use in Class I Division 2 (CID2), Groups A, B, C and D locations or non-hazardous locations only.***

WARNING ***Explosion hazard – Do not connect or disconnect equipment unless power has been switched off or the area is known to be non-hazardous.***

AVERTISSEMENT – Risque d'explosion – Avant de connecter ou de déconnecter l'équipement, couper le courant ou s'assurer que l'emplacement est désigné non dangereux.

WARNING ***Explosion hazard – Substitution of components may impair suitability for Class I, Division 2.***

AVERTISSEMENT – Risque d'explosion – La substitution de composants peut rendre ce matériel inacceptable pour les emplacements de Class I, Division 2.

Shipping Box Contents

ORBCOMM's shipping box contains the following parts.

- ST 6100 CID2 bottom connector terminal on an adapter bracket - Qty 1
- Mating cable connector * - Qty 1
- Dielectric grease - Qty 1
- Conduit adapter - Qty 1
- Strain relief bracket - Qty 1
- Cable ties - Qty 2
- Conduit adapter mounting hardware - Qty 4 sets

* Solution Provider may provide a cable instead of the mating connector.

Recording the Mobile ID

The terminal's mobile ID is located in three places: a label on the bottom of the terminal mounting bracket, a label on the original ORBCOMM packaging box and electronically inside the terminal. The mobile ID is a 15-digit alphanumeric identifier in the format NNNNNNNNSKYXXXX.

This number is a unique identifier for the terminal. Record this number before the terminal is mounted. You will need it later for commissioning the terminal on the satellite network.

Installation Steps

Selection of Mounting Location

Before installing the terminal consider the following important instructions:

- Mount the terminal so that the Inmarsat satellite is within α degrees of the mounting surface, α = 0 to 90 degrees for ST 6100 terminals. Make sure that the line of sight is clear of obstructions.
- Mount the terminal where it has a clear view of the sky/satellite. For a mobile installation, this means at the highest point on the vehicle or vessel where it has a clear view of the sky in all directions.
- Do not mount the terminal close to other electrical equipment (Electronic Magnetic Interference) because it may interact because of radiated and conducted emissions. Do not mount the terminal close to radar or other communication antennas.
- The terminal's mounting surface must be capable of mechanically supporting it. It is recommended that the terminal be mounted on an NEMA Type 4 enclosure qualified for use in hazardous locations as required for the specific application. A Galvanized Rigid Conduit (GRC) or Electrical Metal Tubing (EMT) tubing and/or metal GRC/EMT conduit fitting may be used to adapt the terminal.
- Mount the terminal at least 20 cm (8 in) away from all persons.
- Do not mount the terminal where water may build-up or collect.

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Contact your Solution Provider for full details on mounting location guidelines. Performance of the terminal may be enhanced by pointing the terminal at the satellite using a conduit elbow.

The terminal is supplied with an aluminum conduit adapter with a standard 1¼" female thread. The Installer is responsible for integrating the terminal into an installation that meets the local code requirements for hazardous locations.

Materials and Tools Required

Installing a terminal requires the following materials and tools. These materials do not ship with the terminal because they may differ for each installation.

- ST 6100 cable assembly using the ORBCOMM supplied mating cable connector.

  Note: ***It is recommended you choose a raw cable with the following properties:***

  **** The terminal accepts input ranges of 9 to 32 VDC. If your application requires extended cable lengths, it is necessary to calculate the cable voltage drop to determine if the terminal is receiving at least 9 V (with 1.7 A draw). Large cable voltage drops may adversely affect terminal operation.***

  **** Cable jacket and internal conductor installation rated for minimum temperature range of -40°C to 85°C, and UV compliant where exposure to UV is expected.***

  Note: ***For reliable operation, shield all cables used for power and data connections to the terminal.***

- CID2 hazardous location approved wiring enclosure, conduit, and fittings/connections.

  Note: ***The adaptation of the terminal to conduit or other enclosure must comply with local requirements for hazardous location equipment. At a minimum all connections must maintain NEMA Type 4 protection of the cable route and associate connections.***

- A socket or open-end wrench set

ST 6100 Cable

Although ORBCOMM ships the terminal with a mating connector, the Solution Provider may provide an ST 6100 cable. Contact your Solution Provider if you did not receive a terminal cable, because the instructions for recommended cable assembly construction of this cable are beyond the scope of this guide.

Installation

Once the mounting location has been selected, mount the terminal following the steps as indicated below.

1. Secure the conduit adapter (Figure 1) to the asset using hazardous location approved 1¼" conduit and fittings. An example of appropriate fitting is shown in Figure 1. Use an outdoor grade flexible adhesive sealant at each connection to maintain a NEMA Type 4 watertight, dust tight protection.

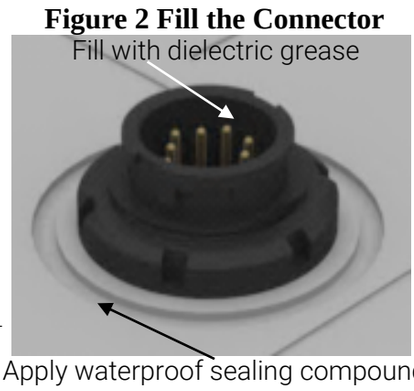
Figure 1 Example Conduit Fitting



2. Pass the cable connector through the conduit and conduit adapter.
3. Cut off one end of the tube of dielectric grease. Adding the dielectric grease ensures that there is a watertight seal.
4. Insert the tube into the male end of the mating connector and squeeze the dielectric grease into the mating connecter opening (Figure 2) until it is full. Some dielectric grease will spill out.

CAUTION ***Do not apply pressure to the cable/connector during the installation.***

Figure 2 Fill the Connector

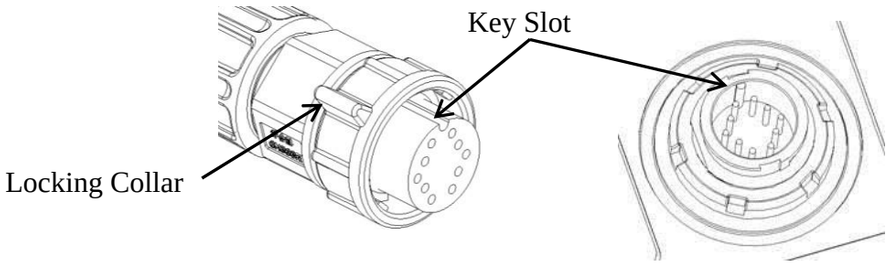


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5. Connect the cable connector to the terminal by aligning the corresponding connector key slot (Figure 3) and gently squeezing together.

CAUTION ***Do not force the connector pins to mate because this may damage the pins.***

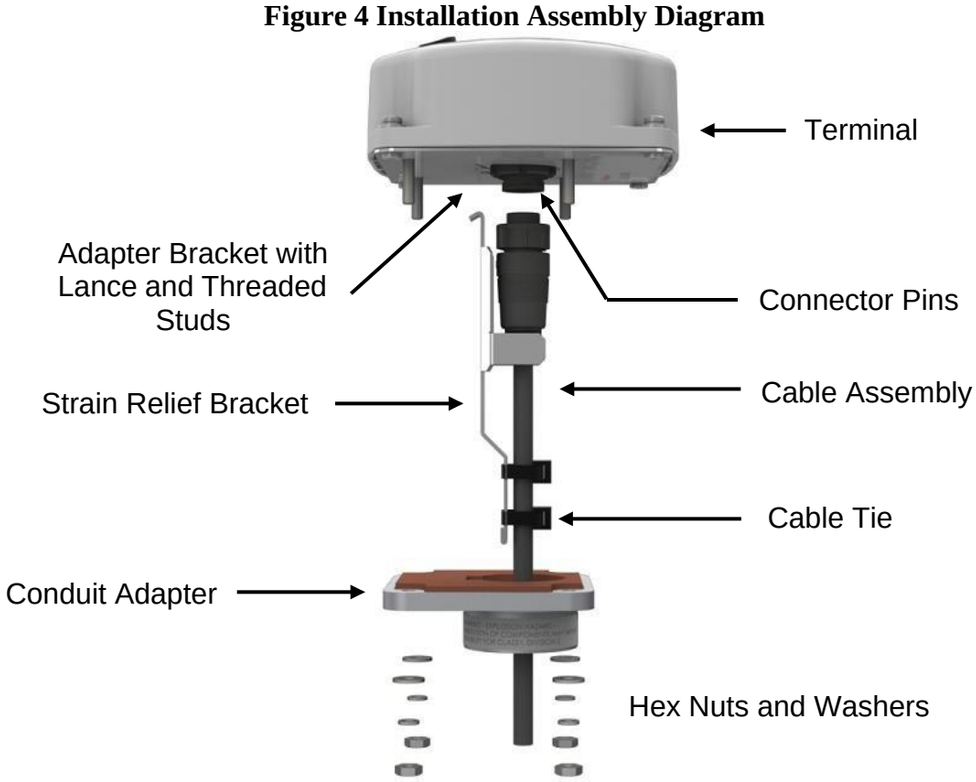
Figure 3 Key Slot



6. Using your fingertips, tighten the cable connector with hand pressure by rotating the locking collar on the cable connector clockwise. A tactile click is felt when the collar is properly engaged. **Do not use your entire hand to rotate the locking collar and do not use a wrench**.
7. Wipe off any extra grease around the connector and wrap the mating connector with waterproof sealing tape if using the connector in changing weather conditions.
8. Position the strain relief bracket in the adapter bracket lance next to the terminal connector.
9. Position two cable ties around the cable at the notches in the strain relief. Do not tighten at this time.
10. Tighten the cable ties while pulling down on the strain relief bracket. This helps ensure that any stress on the cable does not stress the cable/connector joint.
11. Assemble the conduit adapter to the threaded studs in the terminal adapter bracket using four flat washers, lock washers and hex nuts.

Note: ***To improve performance, point the terminal in the direction of the satellite.***

Figure 4 Installation Assembly Diagram



Applying Power

WARNING ***Explosion hazard – Do not connect or disconnect equipment unless power has been switched off or the area is known to be non-hazardous.***

AVERTISSEMENT – Risque d'explosion – Avant de connecter ou de déconnecter l'equipment, couper le courant ou s'assurer que l'emplacement est désigné non dangereux.

Before connecting the terminal to an external voltage source make sure that the polarity is correct, and the voltage source is between 9 and 32 VDC.

Always connect the ground connection at the same time or before power is applied. This is especially applicable to cases when the RS-232 lines are used along with power and ground.

Verification

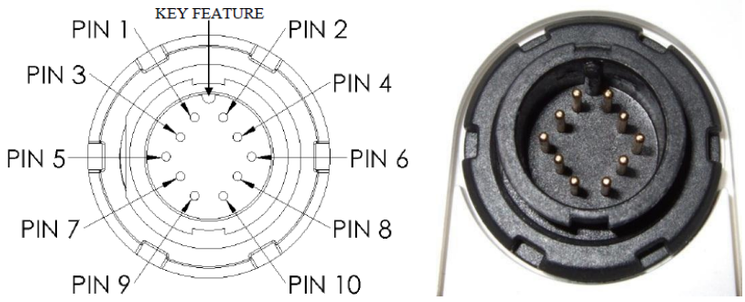
Verification procedures need to be done in cooperation with the Solution Provider as these may vary depending on the installation and the Solution Provider's application.

Appendix Data

Terminal Pin-out for CID2 with RS-485

Pin 1	RS-485 A
Pin 2	RS-485 B
Pin 3	GND
Pin 4	VIN
Pin 5	I/O 02
Pin 6	I/O 03
Pin 7	I/O 01
Pin 8	I/O 04
Pin 9	RS-232 Tx (Output)
Pin 10	RS-232 Rx (Input)

Figure 5 Terminal Connector Pin Assignment (Male Connector)



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